

EMCH 310

Spring 2020 Final Examination

HONOR PLEDGE

I, NAME (print): Alexandra Morrison hereby, state that the solution was obtained by me and I did not seek or receive any help from other students, CHEGG or any online or outside resources. I have neither provided help to anyone else and have maintained the University of South Carolina's Honor Code.

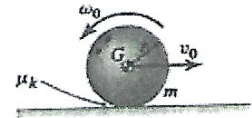
SHOW ALL WORK ON ALL EXAMINATION QUESTIONS TO RECEIVE CREDIT (for the exception of conceptual questions 4, 5, 7, 8, 9, 10, 13, 14, 17 and 18)

You have up to 48 hours to complete and scan the final examination in on BLACKBOARD. Please make sure that you write legibly and scan with good resolution. Convert all of your work to a single pdf file. If Honor Pledge is missing, I will subtract 10 points/100.

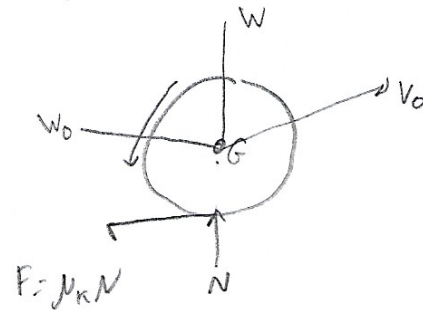
I wasn't in class for when we all are supposed to start the exam so if there were any changes to any variables or questions in the exam, I wasn't there for that.

A bowling ball is thrown onto a lane with a backspin ω_0 , and a forward velocity, v_0 . The mass of the ball m , the radius r , the radius of gyration, k_G , and the coefficient of kinetic friction between the ball and the lane is μ_k . For a 15 lb ball with $r=4.25$ in, $k_G=2.4$ in, $\omega_0=9$ radians/second and $v_0=17$ mph,

- Using "The Principle of Impulse and Momentum" draw the FBD of the ball's motion.
- Determine the time it takes for the ball to start rolling.
- Determine v_2
- Determine the acceleration
- Determine the distance traveled



a)

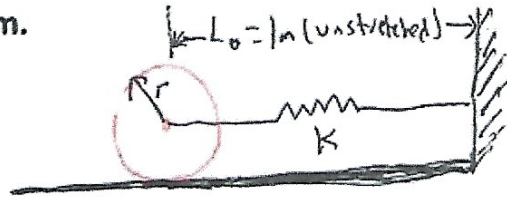


We need the value
of μ_k to work
these problems.

Unless we assume it is zero or keep
as a variable?

A 15 kg thin disk of radius 0.7 m is attached to a spring with an initial length of 1 m. The spring constant $k=25 \text{ N/m}$.

The disk is rolled backwards until the spring is stretched to 3.5 m and then the disk is released from rest. Using "The Principle of Kinetic Energy on Rigid Bodies", determine the velocity in m/s at the center of the disk when the spring length reaches 1 m.



$$\frac{1}{2} k (x_2 - x_1)^2 = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$

$$\frac{1}{2} (25 \text{ N/m}) (3.5 \text{ m} - 1 \text{ m})^2 = \frac{1}{2} (15 \text{ kg}) v^2 + \frac{1}{2} (3.675 \text{ kg} \cdot \text{m}^2) \left(\frac{v^2}{0.7 \text{ m}^2} \right)$$

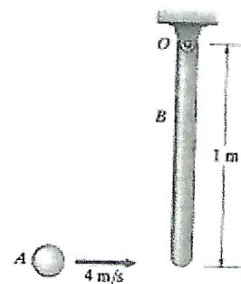
$$78.125 \text{ Nm} = (7.5 \text{ kg}) v^2 + (3.75 \text{ kg}) v^2$$

$$\sqrt{v^2} = \sqrt{6.944}$$

$$v = 2.63 \text{ m/s}$$

$$\text{Ans: } 2.63 \text{ m/s}$$

The 2-kg ball A is moving toward the right at 4 m/s when it strikes the end of the 5-kg slender bar B. The coefficient of restitution is $e = 0.4$. Determine the angular velocity of the rod just after impact.



$$m v_{balli} = m v_{ballf} + m v_{rod} + I \omega_f$$

$$2 \text{ kg} \cdot 4 \text{ m/s} \cdot 1 \text{ m} = 2 \text{ kg} \cdot v_{BF} \cdot 1 \text{ m} + 5 \text{ kg} \cdot v_{rf} \cdot 1 \text{ m} + \frac{1}{3} (5 \text{ kg}) (1 \text{ m})^2 \omega_f$$

$$8 \text{ kg}\cdot\text{m/s} = (2 \text{ kg}\cdot\text{m}) v_{BF} + (5 \text{ kg}\cdot\text{m}) v_{rf} + \frac{1}{3} (5 \text{ kg}) (1 \text{ m})^2 \omega_f$$

$$8 \text{ kg}\cdot\text{m/s} = (2 \text{ kg}\cdot\text{m}) v_{BF} + (5 \text{ kg}\cdot\text{m}) v_{rf} + 1.67 \text{ kg}\cdot\text{m}^2 \cdot \omega_f$$

$$8 \text{ kg}\cdot\text{m/s} = (2 \text{ kg}\cdot\text{m}) v_{BF} + (6.67 \text{ kg}\cdot\text{m}) v_{rf}$$

$$e = \frac{v_{rf} - v_{BF}}{v_{Bi} - v_{ri}}$$

$$v_{Bi} - v_{ri}$$

$$0.4 = \frac{v_{rf} - v_{BF}}{4 \text{ m/s}}$$

$$4 \text{ m/s}$$

$$v_{rf} - v_{BF} = 1.6 \text{ m/s}$$

$$-v_{BF} = 1.6 - v_{rf}$$

$$v_{BF} = v_{rf} - 1.6 \quad (50 \text{ kgs})$$

$$v_{BF} = 1.291 - 1.6$$

$$= 0.308 \text{ m/s}$$

=>

$$8 = 2(v_{rf} - 1.6) + 6.67 v_{rf}$$

$$8 = 2v_{rf} - 3.2 + 6.67$$

$$11.2 = 8.67 v_{rf}$$

$$v_{rf} = 1.2918 \text{ m/s}$$

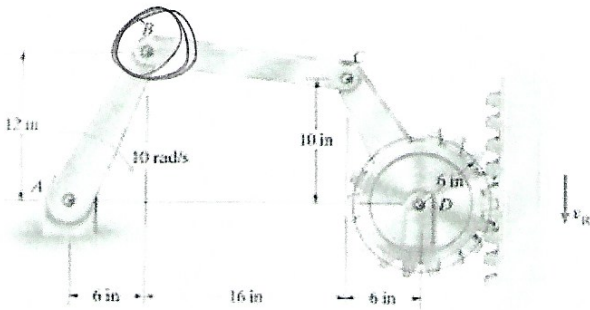
$$8 \text{ kg}\cdot\text{m/s} = (2 \text{ kg}\cdot\text{m}) (-0.308 \text{ m/s}) + (6.67 \text{ kg}\cdot\text{m}) v_{rf}$$

$$8 \text{ kg}\cdot\text{m/s} = -0.616 \text{ kg}\cdot\text{m/s} + (6.67 \text{ kg}\cdot\text{m}) v_{rf}$$

$$= 1.291 \text{ rad/s}$$

21 Conceptual Questions

- 1) Bar AB rotates with a clockwise angular velocity of 10 radians/second.
What is the velocity of point B in m/s?



$$V_D = V_A + \omega_{AB} \times r_{B/A}$$

$$V_D = (6\hat{i} + 12\hat{j}) \times 10\hat{k}$$

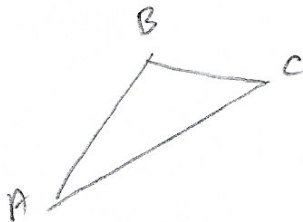
$\hat{i}$	$\hat{j}$	$\hat{k}$
0	0	-10
6	12	0

$$V_D = \sqrt{12^2 + 6^2} = 13.42$$

$$V_D = 10 \text{ rad/s} \times 13.42$$

$$V_D = 134.2 \text{ m/s}$$

- 2) Explain, in words and equations (DO NOT SOLVE) as to how you would get the angular velocity of bar BC.

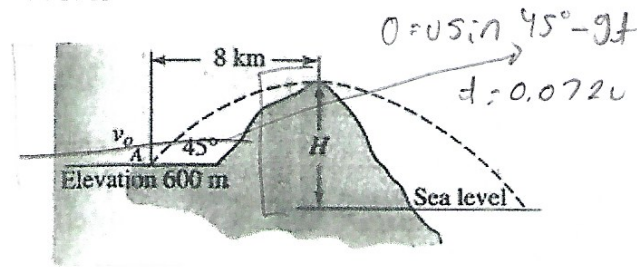


We know the angular velocity
= 10 rad/sec. Since we found
 V_D , we know that V_C can't
move horizontally. You then
use the length of bar
BC at a fixed point.

Separate $\hat{i}$ and $\hat{j}$ and then
solve!

- 3) A long-range artillery rifle located at point A fires a shell at an angle of 45° with respect to the horizontal as illustrated. If the shell just barely clears the mountain peak at the top of its trajectory, the height "H" of the mountain peak above sea level is

- a) 7,998.0 m
- ☒ b) 4,600.0 m
- c) 4,000.0 m
- d) 12,600.0 m



$$v_y = v_{y0} - gt$$

$$x = v_x t$$

$$8000 = v \cos 45^\circ \times 0.072v$$

$$= 4000 + 600$$

- B 4) Which of the following objects has the least kinetic energy at the bottom of an incline if they all have the same mass and radius?

- a) Cylinder
- ☒ b) Sphere
- c) Hoop

- C 5) Which of the following objects has the greatest rotational energy at the bottom of the incline if they all have the same mass and radius?

- a) Cylinder
- b) Sphere
- ☒ c) Hoop

* this question requires radius...? diameter is 20cm but radius is 10cm...

6) A solid cylinder with diameter 20 cm has an angular velocity of 10 radians/second and angular momentum of $2 \text{ kg m}^2/\text{s}$. What is its mass?

- a) 0.1 kg
- b) 1 kg
- c) 5 kg
- ☒ d) 10 kg

$$2 \text{ kg m}^2/\text{s} = \frac{1}{2} m \cdot (0.1 \text{ m})^2 \cdot (10 \text{ rad/s})$$

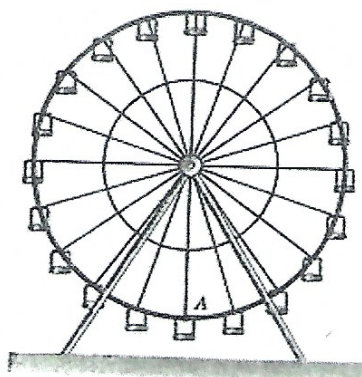
$$2 = 0.4m$$

$$= 40 \text{ kg}$$

if we use diameter rays we get 10

B

7) The Ferris wheel is rotating with a constant angular velocity ω . What is the direction of the acceleration at Point A?

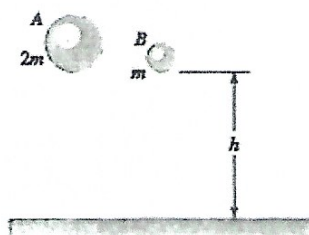


- (a) $\rightarrow$
- ☒ (b) $\uparrow$
- (c) $\downarrow$
- (d) $\leftarrow$
- (e) The acceleration is zero

C

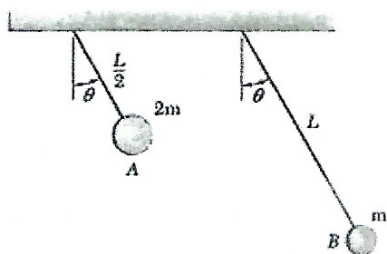
8) Two small balls A and B with masses $2m$ and m are released from rest at height "h" above the ground. Neglecting air resistance, which of the following statements are true when the 2 balls hit the ground?

- a) The K.E. of A is the same as the K.E. of B
- b) The K.E. of A is $\frac{1}{2}$ the K.E. of B
- ☒ c) The K.E. of A is twice the K.E. of B
- d) The K.E. of A is 4 times the K.E. of B



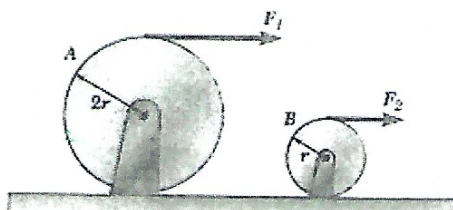
- B 9) Two pendulums A and B with the masses and lengths shown are released from rest. Which system has a larger mass moment of inertia?

a) A
 (b) B



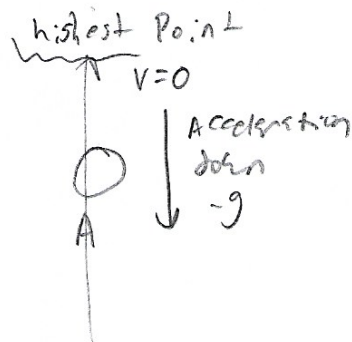
- B 10) Two solid cylinder A and B have the same mass m and the radii $2r$ and r respectively. Each is accelerated from rest with a force applied as shown. In order to impart identical angular accelerations to both cylinders, what is the relationship between F_1 and F_2 ?

a) $F_1 = 0.5F_2$
 (b) $F_1 = 2F_2$
 c) $F_1 = 4F_2$
 d) $F_1 = 8F_2$

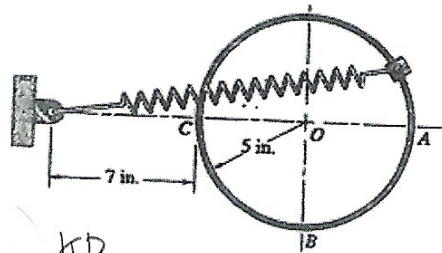
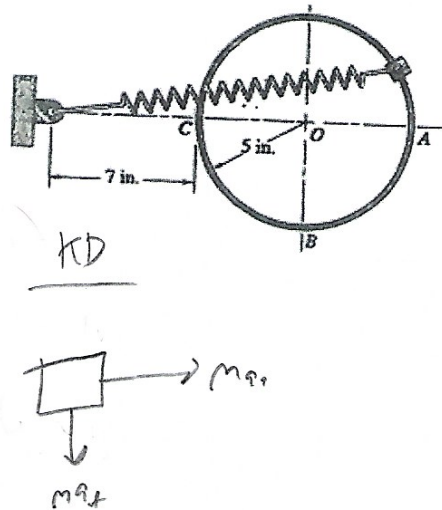
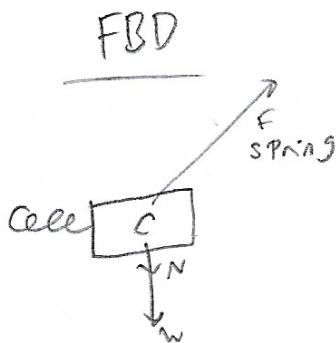


- B 11) Ball A is thrown up. Which of the following statements about the ball are true at the highest point in its path?

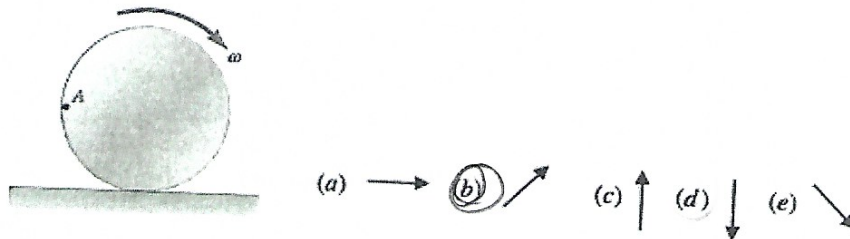
a) The velocity and the acceleration are both 0
 (b) The velocity is 0, but the acceleration is not 0
 c) The velocity is not 0, but the acceleration is 0
 d) Neither the velocity nor the acceleration are 0



- 12) A collar of mass m is attached to a spring and slides without friction along a circular rod in a vertical plane. The spring has an undeformed length of 5 inches and a constant " k ". Knowing that the collar has a speed v at Point C, simply DRAW BOTH THE FBD AND THE KD OF THE COLLAR. DO NOT SOLVE.



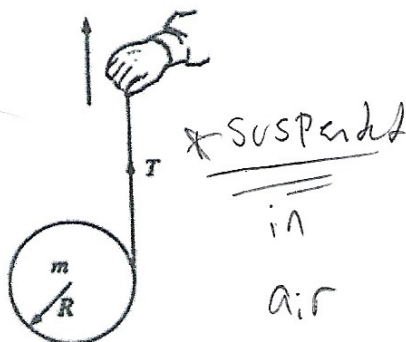
- B 13) The ball rolls without slipping on the fixed surface shown. What is the direction of the velocity at Point A?



- A 14) The measure of the body's resistance to angular acceleration is called _____
- a) Mass moment of inertia of a body
 - b) Volume moment of inertia of a body
 - c) Line moment of inertia of a body
 - d) Gauss moment of inertia of a body

Questions 15 and 16 are based on the following:

A solid cylinder of mass m and radius R has a string wound around it. A person holding the string pulls it vertically upward such that the cylinder is suspended in midair for a brief time interval (Δt) and its center of mass does not move. The tension in the string is " T ", and the rotational inertia of the cylinder about its axis is $\frac{1}{2}mr^2$.



15) The net force on the cylinder during the time interval (Δt) is

- a) T
 b) mg
 c) $T - mgr$
 d) $mgr - T$
 e) zero

$$T - mg = ma$$

$$T - mg = 0$$

$$ma = 0$$

$$F_{\text{net}} = Mc$$

$$F_{\text{net}} = \boxed{0}$$

Suspended

$$T - mg = 0$$

16) The linear acceleration of the person's hand is during the time interval (ΔT) is

- a) $T - mg / m$
 b) $2g$
 c) $g/2$
 d) T/m
 e) Zero

$$a + a_n = \frac{2T}{m}$$

$$T = mg$$

$$a + a_n = 2g$$

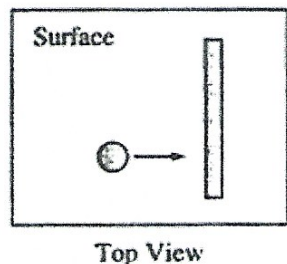
C 17) A figure skater goes into a spin with arms fully extended. Which of the following describes the changes in the rotational kinetic energy and angular momentum of the skater as the skater's arms are brought to the body?

- a) Rotational K.E. remains the same and the angular momentum increases
- b) Rotational K.E. remains the same and the angular momentum remains the same
- c) Rotational K.E. remains increases and the angular momentum remain the same
- d) Rotational K.E. decreases and the angular momentum increases
- e) Rotational K.E. decreases and the angular momentum remain the same

D 18) A disk sliding on a horizontal surface that has negligible friction collides with a rod that is free to move and rotate on the surface as shown below. Which of the following quantities must be the same for the disk-rod system before and after the collision?

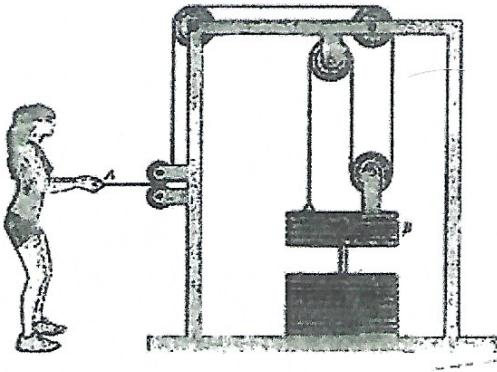
- I. Linear momentum
- II. Angular momentum
- III. Kinetic Energy

- a) I only
- b) II only
- c) I, II and III
- d) I and II only
- e) II and III only



elastic
collisions

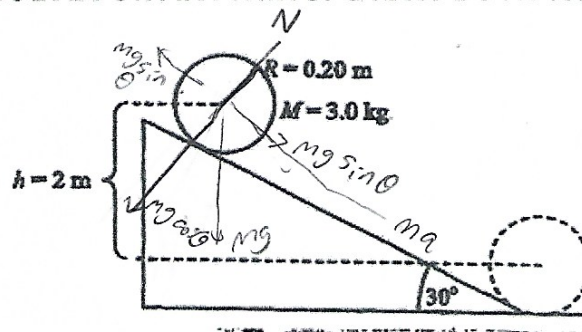
- 19) An athlete pulls handle A to the left with a constant acceleration. Knowing that after the weight B has been lifted 4 inches its velocity is 2 ft/s, WRITE THE DEPENDENT MOTION EQUATIONS AS TO HOW YOU WOULD SOLVE FOR THE ACCELERATION OF HANDLE A. DO NOT SOLVE.



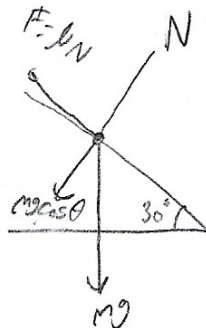
$$(1) V_A = U_A + a_A t$$

$$(2) S_A = U_A t + \frac{1}{2} a_A t^2$$

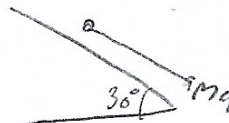
- 20) A spherical mass rolls down an incline plane with friction along the plane. DO NOT SOLVE FOR ANYTHING. DRAW BOTH THE FBD AND KD of the system.



FBD



KD



- 21) Two uniform disk and two cylinders are assembled. Disk A weights 20 lb and disk B weights 12 lb. The system is released from rest. DO NOT SOLVE FOR ANYTHING. DRAW THE FBD AND KD FOR THE ENTIRE SYSTEM.

